

High Level Design (HLD)

HR Analytics

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Mayank Yadav
&
Nitish Kr Dash

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Abstract

The HR Analytics project tackles the critical issue of employee attrition within organizations by employing a data-driven approach to understand and mitigate this challenge. By leveraging a comprehensive dataset of employee records, the project aims to predict attrition rates and identify the key factors influencing employee turnover.

Utilizing advanced machine learning algorithms and statistical analysis techniques, this project analyzes various employee attributes, such as job satisfaction, compensation, tenure, and work-life balance. Through a systematic methodology that includes data preprocessing, exploratory data analysis (EDA), feature engineering, and model development, the project seeks to provide actionable insights for HR professionals.

The primary objective of this project is to empower organizations with the capability to proactively manage employee attrition, ultimately enhancing workforce stability and productivity. The insights derived from the analysis will enable HR departments to make informed decisions aimed at improving employee retention strategies.

1 Introduction

1.1 Why this High-Level Design Document?

The High-Level Design (HLD) document serves as a roadmap for the design, implementation, and deployment of the HR Analytics project. It provides a detailed overview of the system architecture, project goals, scope, and functionality. This document is critical for ensuring that all project stakeholders are aligned and understand how the system will be built to achieve the project's objectives, including predicting employee attrition and identifying key drivers for employee turnover. The HLD outlines all tools, methods, and KPIs necessary to measure the project's success.

The HLD will:

- Present all of the design aspects and define them in detail
- Describe the user interface being implemented
- Describe the hardware and software interfaces
- Describe the performance requirements
- Include design features and the architecture of the project
- List and describe the non-functional attributes

1.2 Scope

The scope of this project includes the end-to-end development of an HR Analytics platform that leverages a historical employee dataset to predict attrition. The main focus is on identifying which factors—such as salary, job satisfaction, work-life balance, and job role—have the greatest impact on whether an employee decides to leave the organization. The project covers:

- **Data Preprocessing:** Cleaning and transforming raw data.
- **Exploratory Data Analysis (EDA):** Understanding trends, correlations, and outliers in the data.
- **Feature Engineering:** Identifying and selecting the most important variables for the model.
- **Model Development:** Training multiple machine learning models to predict employee attrition.
- **Model Evaluation:** Assessing model performance through evaluation metrics.
- **Data Visualization:** Creating dashboards to visually represent trends, key findings, and actionable insights.
- **Deployment:** Making the predictive model and dashboards available for HR professionals.

The ultimate goal is to build a robust system that provides actionable insights to reduce attrition, improve employee satisfaction, and enhance workforce management.

2 General Description

2.1 Product Perspective & Problem Statement

Employee attrition poses a significant challenge to businesses, leading to productivity losses, increased hiring and training costs, and potential damage to employee morale. Understanding why employees leave and predicting who is most likely to leave can help organizations take preventive actions. The aim of this project is to predict employee attrition using a dataset of employee records and identifying the main factors contributing to employee turnover.

The **problem statement** is as follows:

Given historical employee data, predict the likelihood of future attrition for individual employees. Additionally, identify the key factors driving attrition, such as job role, salary, work-life balance, and job satisfaction. These insights will help HR departments proactively implement retention strategies.

2.2 Tools used

To achieve the project objectives, the following tools and technologies will be used:

- **Programming Languages:**
 - **Python:** The primary language for data processing, analysis, and model development due to its rich ecosystem of data science libraries.
 - **R (optional):** For additional statistical analysis if needed.
- **Data Analysis & Visualization Libraries:**
 - **Pandas** and **NumPy:** For data manipulation, cleaning, and numerical computations.
 - **Matplotlib** and **Seaborn:** For creating visualizations during the exploratory data analysis (EDA) phase.
- **Machine Learning Libraries:**
 - **Scikit-learn:** For implementing machine learning algorithms like Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines.
 - **XGBoost** and **LightGBM:** For advanced ensemble methods that provide high performance on structured/tabular data.
 - **CatBoost:** For handling categorical variables efficiently and providing robust models.
- **Data Visualization & Dashboard Tools:**
 - **Tableau:** For creating interactive dashboards that HR professionals can use to monitor employee attrition metrics.
 - **Power BI:** An alternative tool for building visualizations that present insights into employee behavior and attrition trends.

- **Version Control:**
 - **GitHub:** For maintaining the project's codebase, ensuring version control, and enabling collaboration among team members.
- **Deployment Tools:**
 - **Tableau Public** or **Power BI** for deploying the interactive dashboards.
 - **GitHub Pages** (optional): For hosting documentation and sharing project reports and analysis.



NumPy



pandas



Excel



+ a b l e a u ®

3 Design Details

3.1 Functional Architecture

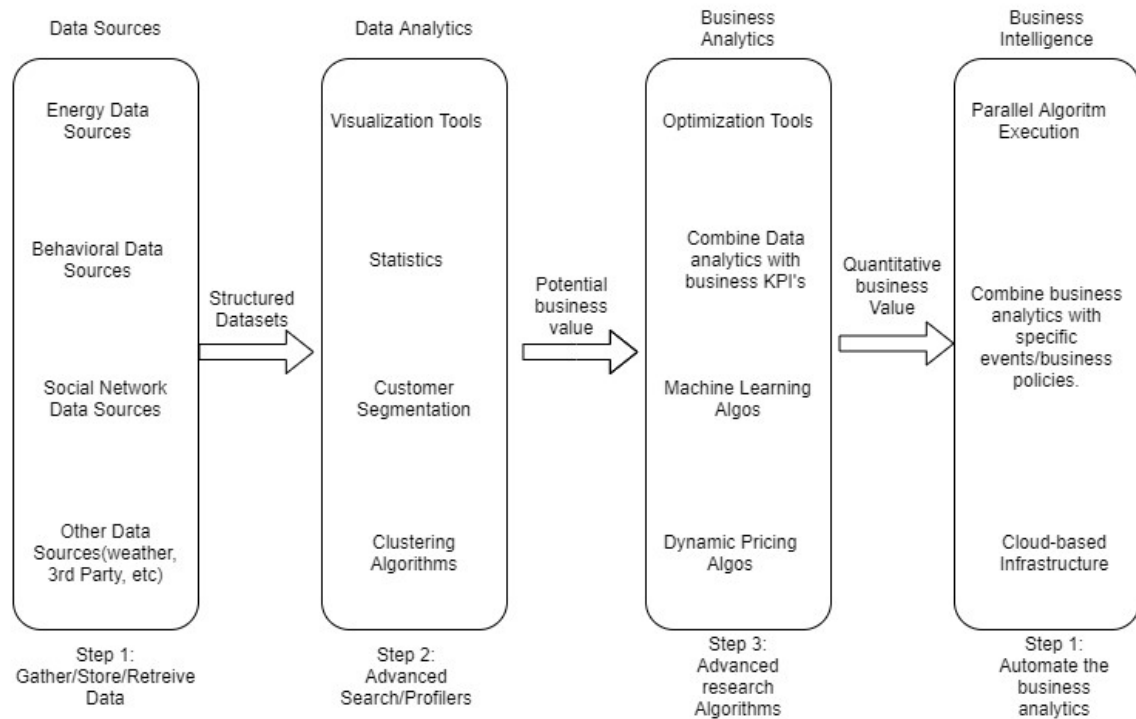
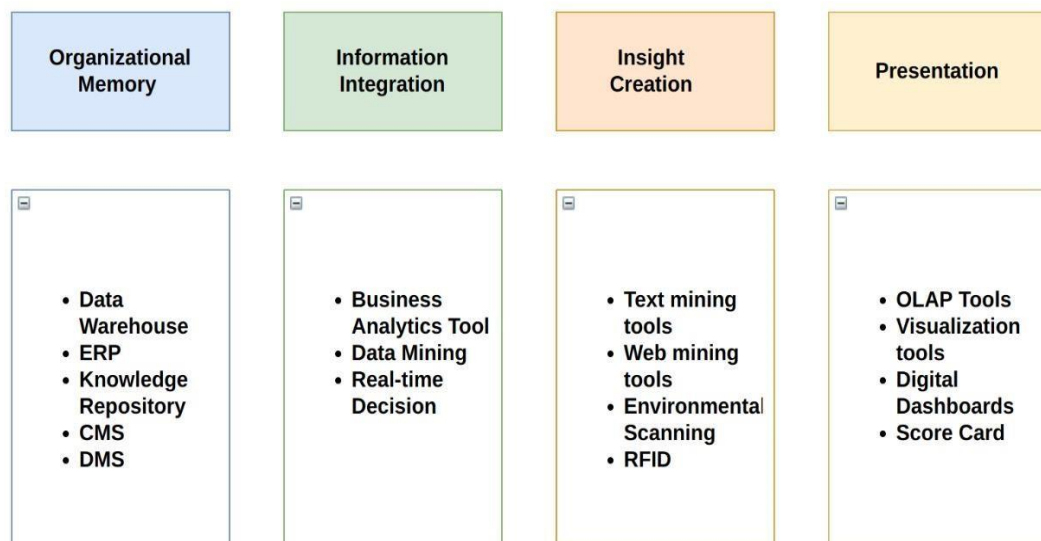


Figure 1: Functional Architecture of Business Intelligence

How BI Really Works



3.2 Optimization

☐ Your data strategy drives performance

- Minimize the number of fields
- Minimize the number of records
- Optimize extracts to speed up future queries by materializing calculations, removing columns and the use of accelerated views

☐ **Feature Selection:** Using feature importance scores from models such as Random Forest and XGBoost, the most relevant features are retained, which helps improve model performance.

☐ **Hyperparameter Tuning:** Model parameters are fine-tuned using techniques such as grid search and random search to achieve the best possible performance.

☐ **Cross-Validation:** K-fold cross-validation ensures that the models generalize well to unseen data and that the performance metrics are robust and reliable.

4 KPIs

4.1 KPIs (Key Performance Indicators)

The following KPIs will measure the effectiveness of the predictive model and the overall success of the HR Analytics project:

- **Model Accuracy:** The percentage of correctly predicted employee attrition cases, indicating how well the model performs.
- **Precision & Recall:** Precision measures how many of the predicted positives (employees likely to leave) were correct, while recall measures how well the model identified all actual positives.
- **ROC-AUC Score:** The area under the ROC curve will provide insight into the model's ability to distinguish between employees who will stay and those who will leave.
- **Attrition Reduction Rate:** This measures the effectiveness of the HR department's interventions in reducing employee attrition.

5 Deployment

1. Environment Setup

The project relies on Python and Jupyter Notebook as the main environment for conducting data analysis and building machine learning models. Ensure that the necessary tools are available and configured correctly.

- **Python Environment:** Ensure Python is set up with all required libraries and dependencies.
- **Jupyter Notebook:** The analysis will be conducted in Jupyter Notebook, where users can interact with the code, execute it, and view the outputs (such as visualizations and model results).

2. Jupyter Notebook Organization

The project files should be well-structured to maintain clarity and ease of use:

- **Data Folder:** Store all datasets (e.g., employee data) in a clearly labeled folder.
- **Notebooks:** The Jupyter Notebooks used for data preprocessing, analysis, and visualizations should be well-documented, with clear headings, markdown cells explaining each section, and clean outputs.
- **Models:** Store the trained machine learning models in a separate folder for easy access and reuse.
- **Documentation:** Include a README file with instructions for using the notebook, explaining each step of the process (data loading, analysis, visualization, and model prediction).

3. Visualizations and Reporting

The project includes visualizations generated using Matplotlib and Seaborn. During the deployment process:

- Ensure that all charts, graphs, and visualizations are clearly labeled with appropriate titles, legends, and axes labels.
- Key insights, such as trends in job satisfaction, tenure, and their relationship with attrition, should be highlighted with visual representations.
- Important visualizations (e.g., heatmaps, distribution plots) should be exported or embedded within reports and presentations for further sharing with stakeholders.

4. Model Deployment

After the machine learning models have been trained, they can be deployed in the following ways:

- **Local Deployment:** The trained models are saved and can be loaded in Jupyter Notebook for generating predictions. This allows HR teams to input new data and get predictions regarding employee attrition.
- **Cloud Deployment (Optional):** If you require a more scalable solution, the model can be deployed to a cloud platform (e.g., AWS, Google Cloud, or Microsoft Azure) to serve predictions for larger datasets or integrate with an existing HR management system.

5. Exporting the Notebook

To share the analysis results with non-technical stakeholders, you can export the Jupyter Notebook in user-friendly formats:

- **HTML Format:** Export the notebook to HTML, allowing viewers to open it in a web browser without needing to run the notebook themselves. This is useful for presenting the analysis and visualizations in a clear and interactive way.

- **PDF Format:** Export the notebook as a PDF for distribution via email or printing. Ensure that all relevant outputs and visualizations are included in the exported file.

6. Version Control & Collaboration

The project should be maintained using version control, especially if multiple team members are collaborating:

- **GitHub Repository:** The project should be stored in a public or private GitHub repository, where collaborators can track changes, update files, and ensure the project is consistently documented. The repository should include the notebook, models, data, and all necessary documentation.
- **Branching Strategy:** Use branches to experiment with different models or analysis techniques without affecting the main project. This ensures that the core project remains stable while allowing innovation and experimentation.

7. Dashboard Deployment

To ensure that the insights gained from the analysis are accessible to HR professionals, interactive dashboards can be created and deployed:

- **Tableau Dashboard:** After analyzing employee attrition, create interactive dashboards in Tableau that summarize key insights such as employee satisfaction levels, performance, and factors influencing attrition. Once the dashboards are complete, publish them to **Tableau Public** to allow stakeholders to explore the data.
- **Power BI Dashboard:** Similarly, you can use Power BI to build interactive dashboards, visualizing important employee trends and attrition risks. These dashboards can be published to **Power BI Service**, where they can be shared with decision-makers.

8. Continuous Monitoring and Updates

Since employee attrition is an ongoing issue, the model and analysis will require continuous updates as new data becomes available:

- **Data Updates:** Regularly update the dataset with new employee records to ensure that the model predictions and insights remain accurate and relevant.
- **Model Retraining:** Retrain the model periodically to account for new trends in employee behavior or changes in business strategy.
- **Dashboard Updates:** Ensure that the visualizations and dashboards are refreshed with the latest data, providing HR teams with up-to-date information.

Deployment Checklist

- Jupyter Notebook organized with well-documented sections and outputs.
- Visualizations embedded clearly and saved for reporting.
- Machine learning models saved for future prediction.
- Notebooks exported to HTML or PDF for easy sharing.
- GitHub repository set up for version control.
- Dashboards deployed via Tableau or Power BI for HR professionals.
- Plan for continuous monitoring and updating of data and models.